

14. Segmentation U-Net Mask R-CNN

Overview

Semantic segmentation assigns a class per pixel; instance segmentation separates individual objects. U-Net skip connections preserve spatial detail; Mask R-CNN extends Faster R-CNN with a parallel mask head. Losses, metrics (mIoU, Dice), and augmentation for small datasets are covered.

Lab: Lab 14.

Prior modules: Module 12, Module 13.

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Module track

This module builds on **Module 12**, **Module 13**. Read those PDFs first. References to other modules appear as **Module NN** (not page numbers, because each module is its own PDF).

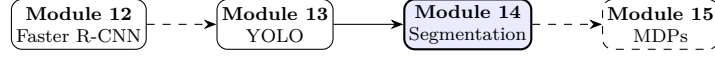


Figure 1: Module sequence (solid box: this PDF; dashed: typical next step).

1 Dense Prediction Tasks

Segmentation assigns a class label to every pixel. Architectures extend detectors (**Module 12** (Faster R-CNN pipeline)) and encoder to decoder designs.

2 U-Net Architecture

U-Net [5] uses contracting and expanding paths with skip connections:

$$\mathbf{y} = \text{Decoder}(\mathbf{z}, \{\mathbf{s}_\ell\}) = \text{softmax}(\hat{\mathbf{y}}), \quad \mathbf{z} = \text{Encoder}(\mathbf{x}). \quad (1)$$

Skip features $\mathbf{s}_\ell$ preserve fine spatial detail lost in pooling.

Note. Skip connections mirror the denoising U-Net in **Module 08**, shared inductive bias for spatial reconstruction.

3 U-Net Loss

Weighted cross-entropy handles class imbalance in biomedical masks:

$$\mathcal{L} = - \sum_p w_p \sum_c y_{p,c} \log \hat{y}_{p,c}. \quad (2)$$

Dice loss alternative: $\mathcal{L}_{\text{Dice}} = 1 - 2|X \cap Y|/(|X| + |Y|)$.

4 Mask R-CNN

Extends Faster R-CNN (**Module 12**) with parallel mask head per RoI:

$$\mathcal{L} = \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{mask}}. \quad (3)$$

Mask branch outputs $m \times m$ binary mask per class (e.g. 28×28), upsampled to full resolution.

5 Instance vs. Semantic Segmentation

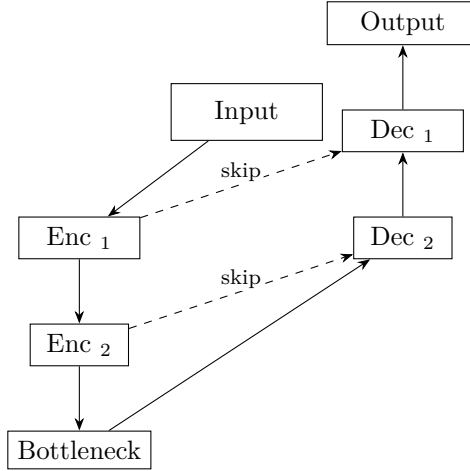
Semantic: one label per pixel. Instance: separate objects of same class. Panoptic combines both.

6 Evaluation Metrics

$$\text{IoU}_{\text{mask}} = \frac{|M \cap M^{\text{gt}}|}{|M \cup M^{\text{gt}}|}, \quad \text{mAP}^{\text{mask}} = \frac{1}{C} \sum_c AP_c^{\text{mask}}. \quad (4)$$

Table 1: Segmentation method comparison.

Method	Instance	Arbitrary classes	Speed
U-Net	No	Yes	Fast
Mask R-CNN	Yes	Yes	Moderate
DeepLab	No	Yes	Moderate

**Figure 2:** U-Net hourglass with lateral skip connections.

7 Lab

Lab 14 trains U-Net on semantic masks; optional Mask R-CNN extension using RoI heads from **Module 12**.

8 DeepLab and Atrous Conv

Atrous spatial pyramid pooling captures multi-scale context:

$$\mathbf{y} = \text{Concat}(\text{Conv}_{d=1}\mathbf{x}, \text{Conv}_{d=6}\mathbf{x}, \text{Conv}_{d=12}\mathbf{x}). \quad (5)$$

COCO panoptic quality $\text{PQ} = \text{RQ} \times \text{SQ}$ unifies detection and segmentation.

8.1 Data Augmentation

Random scale, flip, and color jitter improve U-Net generalization on small medical datasets [5].

9 Extended Intuition

Semantic segmentation assigns a class label to every pixel; instance segmentation additionally separates individual object instances. U-Net uses an encoder-decoder with skip connections so high-resolution spatial detail lost in downsampling is restored for pixel-wise predictions. Mask R-CNN extends Faster R-CNN (**Module 12**) with a parallel mask head per RoI that outputs a 28×28 mask per instance.

Loss combines per-pixel cross-entropy (or Dice) for semantics with detection losses in instance setups. Metrics: mean IoU (mIoU) averages class IoU across classes; Dice coefficient $= 2|A \cap$

$B/|A| + |B|$ common in medical imaging. Data augmentation (flips, color jitter, random scale) matters more than architecture tweaks on datasets under 500 images.

For multi-class semantic maps, ignore index 255 (or a dedicated void class) in loss for unlabeled pixels. Boundary errors often dominate visually even when mIoU looks acceptable; inspect failure cases on thin structures.

10 Numerical Walkthrough

U-Net on 256×256 RGB, 3 classes (background, road, vehicle), $B = 4$, base channels 64.

Encoder path: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$ spatial with doubling channels. Decoder upsamples and concatenates skip features; final logits (B, 3, 256, 256).

One pixel with logits $[2.0, 1.0, 0.5]$ and target class 0: CE loss = $-\log \text{softmax}([2, 1, 0.5])_0 \approx 0.41$. Batch mIoU: per class compute IoU on valid pixels, average across classes with present GT.

Training 200 epochs on 150 images (small): lr = 10^{-3} Adam, weight decay 10^{-4} , may reach mIoU 0.65 to 0.75 depending on labels. Mask R-CNN adds mask loss ≈ 0.1 to 0.3 alongside box/class losses when fine-tuning on COCO subsets.

11 PyTorch Implementation Notes

- `segmentation_models_pytorch` or custom U-Net: encoder `resnet34`, `classes=3`, `encoder_weights="imagenet"`.
- Masks shape (B, H, W) long dtype with class indices, or (B, C, H, W) one-hot for multi-label.
- `nn.CrossEntropyLoss(ignore_index=255)` for semantic segmentation.
- `torchvision.models.detection.maskrcnn_resnet50_fpn` for instance masks; masks in target are `UInt8Tensor` (N, H, W) per object.
- Upsample logits to label resolution with `F.interpolate(..., mode="bilinear", align_corners=False)` if sizes mismatch.

```
logits = unet(images) # (B, C, H, W)
loss = F.cross_entropy(logits, masks, ignore_index=255)
pred = logits.argmax(dim=1)
```

12 Local GPU Constraints

U-Net 256×256 , $B = 4$, ResNet34 encoder fits 4 GB comfortably; 512×512 needs $B = 2$. Mask R-CNN similar to Faster R-CNN: $B = 1$ to 2 at 800px on 1650.

Use mixed precision for U-Net training; watch `align_corners` consistency between train and eval interpolation. Patch-based training (random 128 crops) simulates larger batches on memory-limited GPUs.

13 Debugging Checklist

- mIoU near zero: all-background predictions; check class imbalance and loss ignore index.
- Checkerboard in upsampling: transposed conv artifacts; prefer bilinear upsample + conv.
- Mask R-CNN mask all zeros: mask targets not aligned to box coordinates after resize.
- OOM at 512px: reduce batch or use lighter encoder (`mobilenet_v2`).
- Val mIoU noisy on tiny val set: merge val images or use sliding-window inference.

13.1 Mask Heads, Class Imbalance, and Inference Modes

Semantic segmentation treats every pixel independently after the decoder; instance segmentation (Mask R-CNN) predicts one soft mask per detected instance inside each RoI. Mask R-CNN mask logits have shape 28×28 per RoI; they are upsampled to image resolution and binarized at 0.5 during evaluation. RoI Align uses bilinear interpolation at continuous bin locations, fixing the quantization error that RoI Pooling introduced; always prefer Align for mask quality on thin structures. Class imbalance (95% background pixels) makes raw accuracy misleading; track per-class IoU and frequency-weighted mIoU so rare classes are not hidden.

Weighted CE loss with class weights inversely proportional to pixel frequency helps road/vehicle splits on aerial imagery. Dice loss optimizes overlap directly and pairs well with CE in a combined objective $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{Dice}}$ with $\lambda \in [0.5, 1.0]$. For medical volumes, 3D U-Net extends the same encoder-decoder pattern; the lab focuses on 2D but the skip-connection rationale is identical.

Inference modes: whole-image forward for 256×256 fits in memory; at 1024+ use sliding windows with overlap (stride 128 on 256 patches) and average logits in overlap regions. Test-time augmentation (horizontal flip, average logits) adds $2\times$ compute but can recover 1 to 2 mIoU points on boundary-heavy classes. Post-processing: remove small connected components below 100 pixels for semantic maps; for instance masks, keep components tied to detection scores above threshold. Metrics code must apply `ignore_index` consistently in val and test; a single off-by-one void label can drop mIoU by 20 points while loss looks fine. Segmentation checkpoints pair naturally with detection outputs from **Module 12** and **Module 13** in multi-stage pipelines (detect then segment inside box). DeepLab-style atrous convolutions widen receptive field without extra downsampling; useful when objects span large image fractions and U-Net depth is limited by VRAM. Panoptic segmentation unifies semantic and instance labels; industrial labs often start with semantic-only U-Net before adding Mask R-CNN complexity. Label noise at object boundaries (± 2 pixel shifts) can drop mIoU 5 to 10 points; audit 20 random overlays in color before blaming the architecture. Color jitter augmentation helps outdoor scenes; disable heavy hue shifts on medical imaging where tissue color carries diagnostic signal. Encoder depth vs decoder depth: shallow encoder (ResNet18) trains faster on 1650 but may miss context; ResNet34 is the lab sweet spot at 256×256 . Binary segmentation can use single-channel logits + `BCEWithLogitsLoss` instead of 2-class CE; numerically stable when foreground is under 5% of pixels. Instance mask visualization: overlay semi-transparent color per instance id; verify mask edges align with bbox from **Module 12** before trusting joint metrics. Focal loss variant $-(1 - p_t)^\gamma \log p_t$ down-weights easy pixels; try $\gamma = 2$ when background dominates and CE plateaus with high train mIoU. Crisp boundary metrics: track Boundary IoU (BIOU) on a 10-image subset; mIoU can hide 3-pixel shifts that break downstream geometry pipelines. Save predicted masks as PNG palette images for quick human review; grayscale class ids are easy to misread under compression artifacts. Albumentations `CoarseDropout` simulates occlusions; dropped regions must use `ignore_index` in loss, not background class. Polygons vs raster masks: rasterize vertices with consistent fill rules before training or IoU disagrees with annotator exports. Multi-label scenes (water + road co-occurring) need sigmoid per channel, not mutually exclusive single-class CE. Log train and val mIoU every epoch to CSV; segmentation curves are noisier than classification accuracy and need moving averages to spot real trends.

14 Practice Problems

Problem 1. Compute IoU for binary masks that overlap on 100 pixels, union 250 pixels.

Problem 2. Implement Dice loss for binary segmentation and compare to CE on imbalanced data.

Problem 3. Train U-Net on a 3-class road scene set; report per-class IoU.

Problem 4. Optional: add Mask R-CNN mask head visualization for 5 validation images. Complete **Lab 14**; uses RoI ideas from **Module 12**; detection metrics from **Module 13**.

Source certification

This module lists where each core definition, equation, figure, and summary table comes from. Pedagogical simplifications (lab batch sizes, epoch counts, illustrative plots) are labeled in extension sections and are not claimed as optimal production settings.

Table 3: Certified topic map for Module 14.

Topic	Primary source	Where in this PDF
U-Net architecture	[5]	Section 2
Mask R-CNN instance masks	[2]	Section 4
FCN semantic segmentation	[3]	Section 5
DeepLab context	[1]	Extension section

Full bibliographic entries appear below. Figures are schematic unless a caption cites a specific paper figure. If a statement is not listed here, treat it as general ML practice or lab-specific guidance, not a cited theorem.

References

- [1] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, et al. DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. In *IEEE TPAMI*, 2018.
- [2] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *ICCV*, 2017.
- [3] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *CVPR*, 2015.
- [4] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *NeurIPS*, 2015.
- [5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015.